5J Index laws for multiplication and division

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|---|---|
| Learning Objective | To be able to evaluate and simplify exponent forms through the use of index laws. |
| Today We Are | Multiplying and dividing numbers in index forms. |
| Indicators of Progress | • I can multiply and divide numbers in index form |

<u>Recall Activity</u>

**Expand** the following:

$$6a\ (\ 2 + 3a)$$

**Factorise** the following:

$$4a + 6ab + 8ac$$

<u>Lets have a look at Index Forms:</u>

- 3 squared = 3 x 3 = $3^2$
- 3 cubed = 3 x 3 x 3 = $3^3$

What would the index form look like if I had five 3's?

3 x 3 x 3 x 3 x 3 =

Index / Exponent / Power

Base

$$5^6$$

$$4^{2+3} = 4^5$$

$$= 2^{2+3+4+1}$$

$$= 2^{10}$$

$3 \times 3 \times \cancel{3} \times \cancel{3} \times \cancel{3}$

$\cancel{3} \times \cancel{3} \times \cancel{3}$

$3^2$

$3^{5-3} = 3^2$

$7^{6-3} = 7^3$

$6^{8-5} = 6^2$

# Fluency

g h i j

# Problem solving